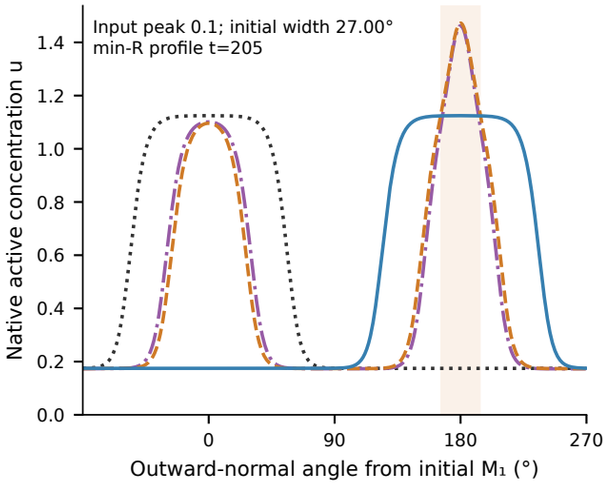
Stationary cues · spatial activity builds a complex vector

$$M_1(t) = \sum_j u_j(t)\,\Delta s_j(t)\,e^{i\theta_j}, \qquad \Delta s_j(t) = C(t)/N, \qquad P(t) = M_1(t)/M_1(0)$$

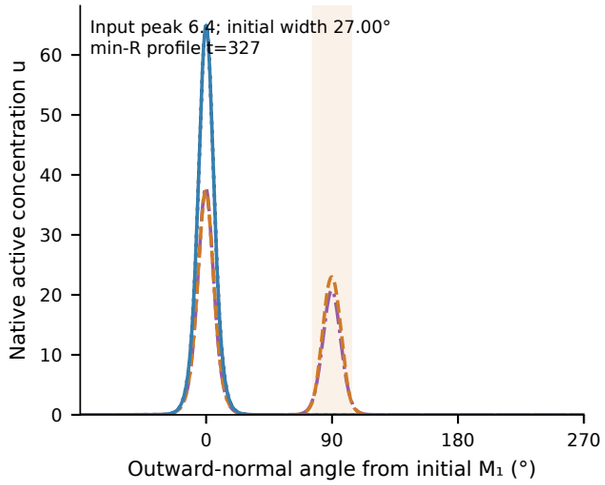
Each contribution is active concentration × physical segment length × outward normal; opposite contributions can cancel. $M_0 = \Sigma u_j \Delta s_j$.

Four actual conditions from the same 80-case experiment · cue fixed in place, then 13,000 cue-free native units · native amplitudes and clocks differ

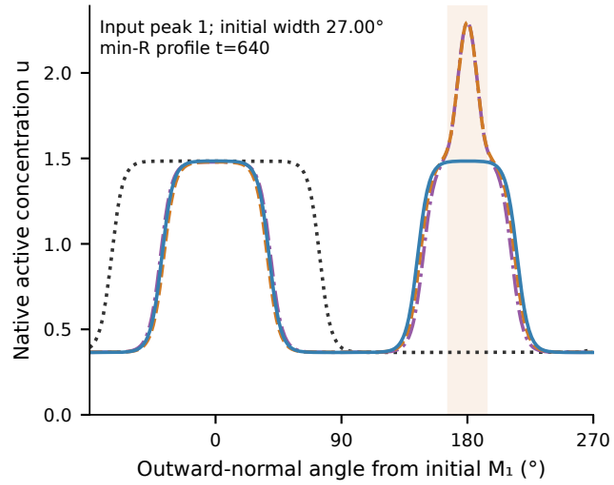
A1 Wave-pinning · 180°, T=300
Retained reversal



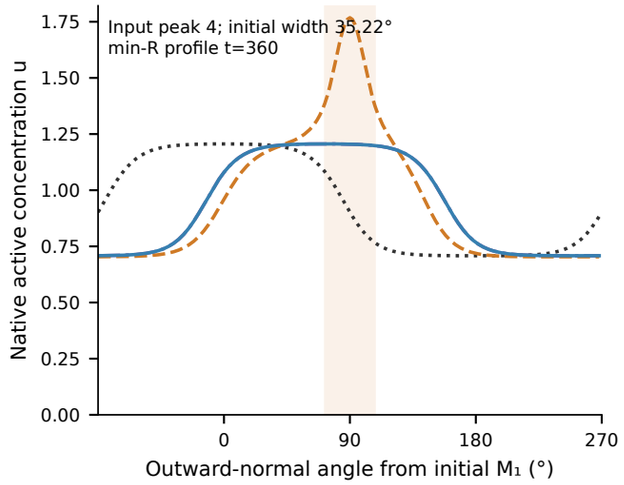
A2 Goryachev · 90°, T=300
Cue excursion and return



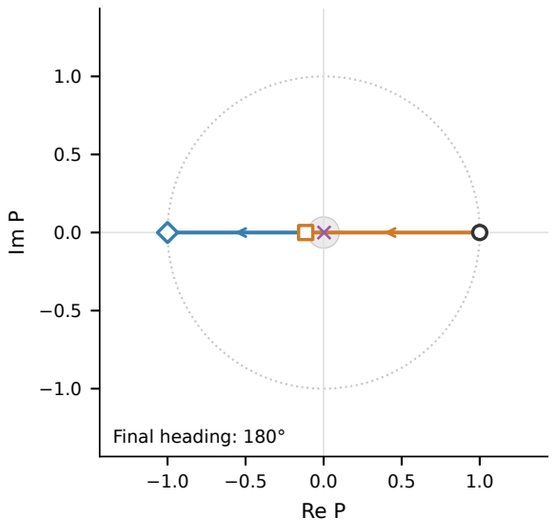
A3 Holmes 3 · 180°, T=1000
Two peaks, weak resultant



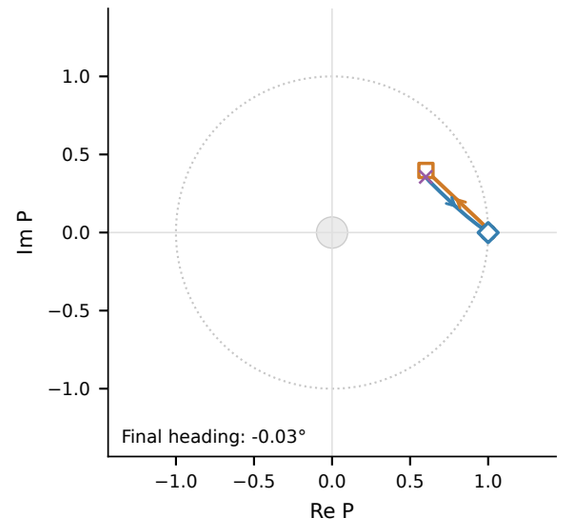
A4 Spring ring · 90°, T=10
Retained graded turn



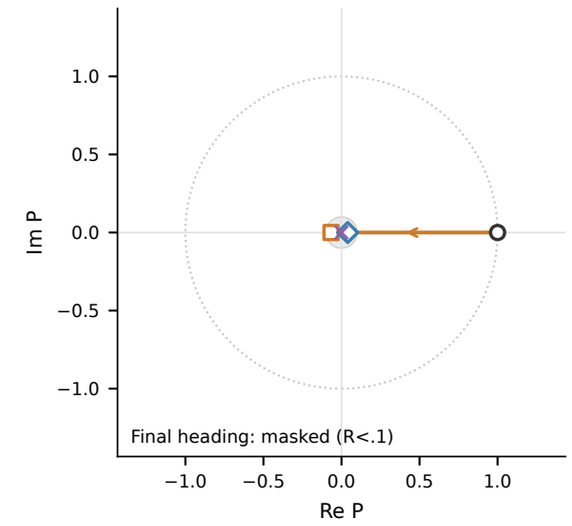
B1 Physical activity-vector trajectory



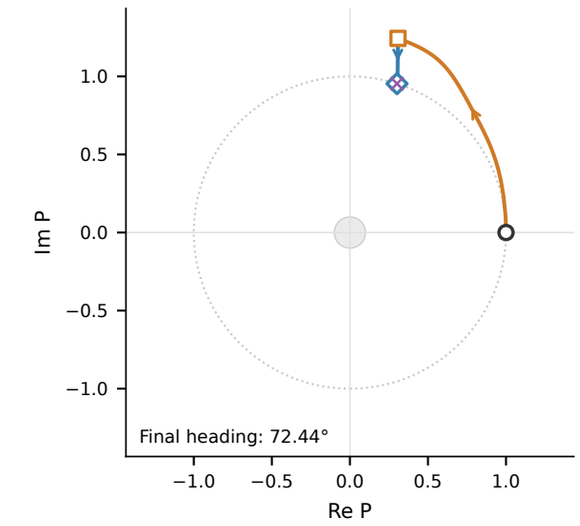
B2 Physical activity-vector trajectory



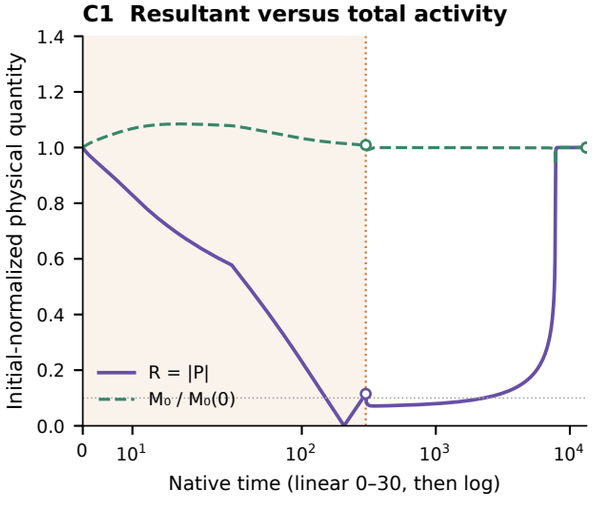
B3 Physical activity-vector trajectory



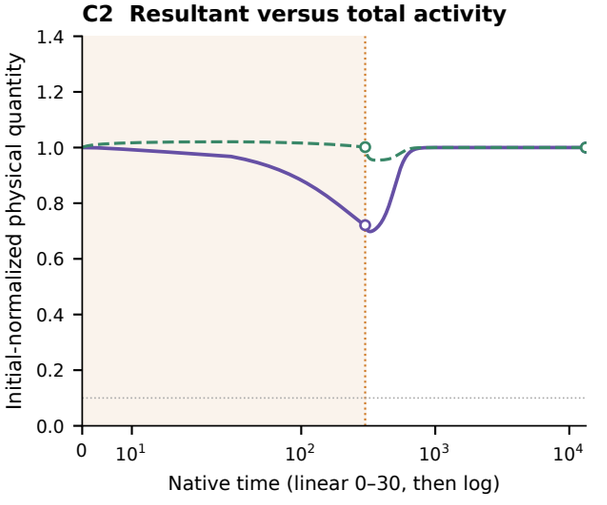
B4 Physical activity-vector trajectory



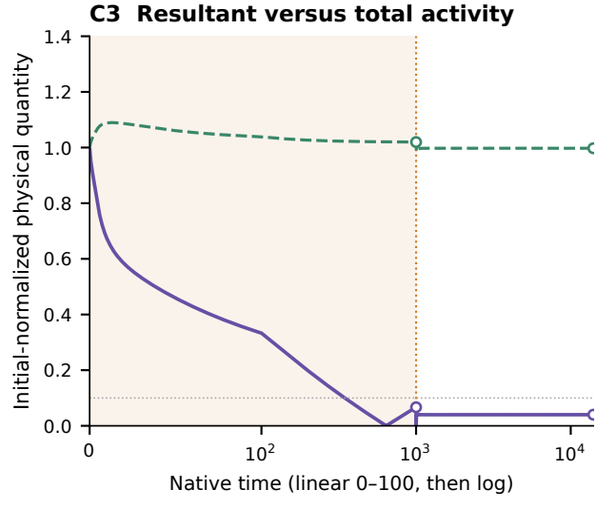
C1 Resultant versus total activity



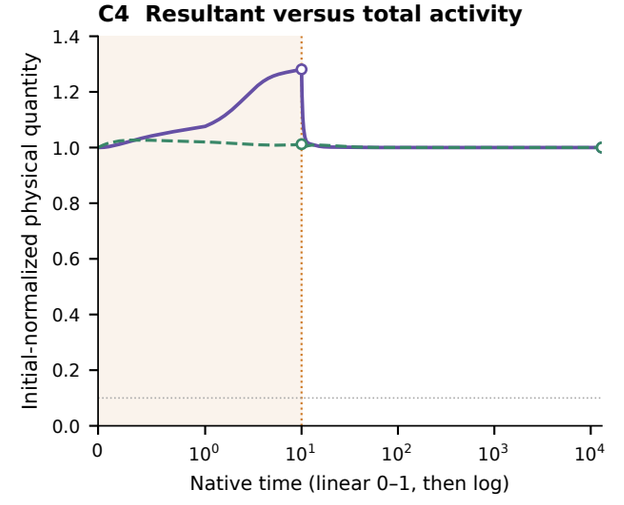
C2 Resultant versus total activity



C3 Resultant versus total activity



C4 Resultant versus total activity



P planes have equal real/imaginary scales and a unit-circle reference. Shaded R<.1 disk: heading is not interpreted. Profile shading: initial cue footprint.

Initial profile · Sampled minimum-R profile · Withdrawal profile · Final profile · Cue on · Cue released · Initial P · Withdrawal P · Final P